TEAM

SkyMate

AGENTIC FLIGHT ASSISTANT · DEL → BOM → ANYWHERE

MEMBER 01	MEMBER 02	MEMBER 03	MEMBER 04	MEMBER 05
Member name	Member name	Member name	Member name	Member name

MENTOR Mentor name

PROBLEM

Planning one flight still takes five tabs and a leap of faith

01 **Scattered information.** Schedules, airline rules, terminals, weather and documents live in different places.

02 **Untrustworthy answers.** Generic chatbots invent fares, gates and timings with no fact-check.

03 **No direct flight, no answer.** Finding connections takes many searches and costs API calls.

04 **No human control.** Costly or irreversible steps, like booking, run without the traveller's go-ahead.

05 **Locked to one use case.** Moving to a new domain means rebuilding the assistant from scratch.

SOLUTION

One chat that plans, checks and asks before it acts

Full walkthrough in the live demo

CHAT	Understands plain requests and searches live flights worldwide, including one-stop connections.
AGENTS	A 12-agent team led by a supervisor builds the itinerary: 9 specialists in parallel, a web agent fills gaps.
SAFETY	Guardrails and fact-checks block injection attempts and check every answer against real data.
HUMAN	Clearances queue for approving connection searches, itinerary reviews and bookings, or saving drafts.
TRIP	Your documents, live weather, SkyPay checkout, calendar and PDF in one place.
GENERIC	Domain packs: change one folder to reuse the whole engine for a new use case.

TECH STACK

Built on LangGraph, LangChain and live data

12	16	3	33	1
agents per itinerary	orchestrated graph steps	human checkpoints	REST endpoints	folder per domain

AI models

via the company LLM gateway

GPT-5-mini · itineraries
GPT-4.1-mini · chat, agents
text-embedding-3-large · RAG
Whisper · voice notes

Orchestration

two supervised graphs

LangGraph 0.4.3
StateGraph · Send
interrupt · Command
SQLite checkpointer 2.0.10

LangChain

model calls and documents

langchain-core 1.6.9
langchain-openai 1.7.0
text-splitters · RAG chunks

Safety & control

trust built in

Rule guardrails
LLM safety classifier
Grounding fact-check
Clearances queue

Data & memory

stored locally

ChromaDB 1.0.9 · vectors
SQLite · approvals, runs
JSON + bcrypt · users

Live data

real-world facts

AeroDataBox · schedules
Open-Meteo · weather
Tavily → Wikipedia · web search

Backend

Python 3.12

Flask 3.1.3
Flasgger · Swagger docs
JWT (PyJWT) · auth
fpdf2 · trip PDF

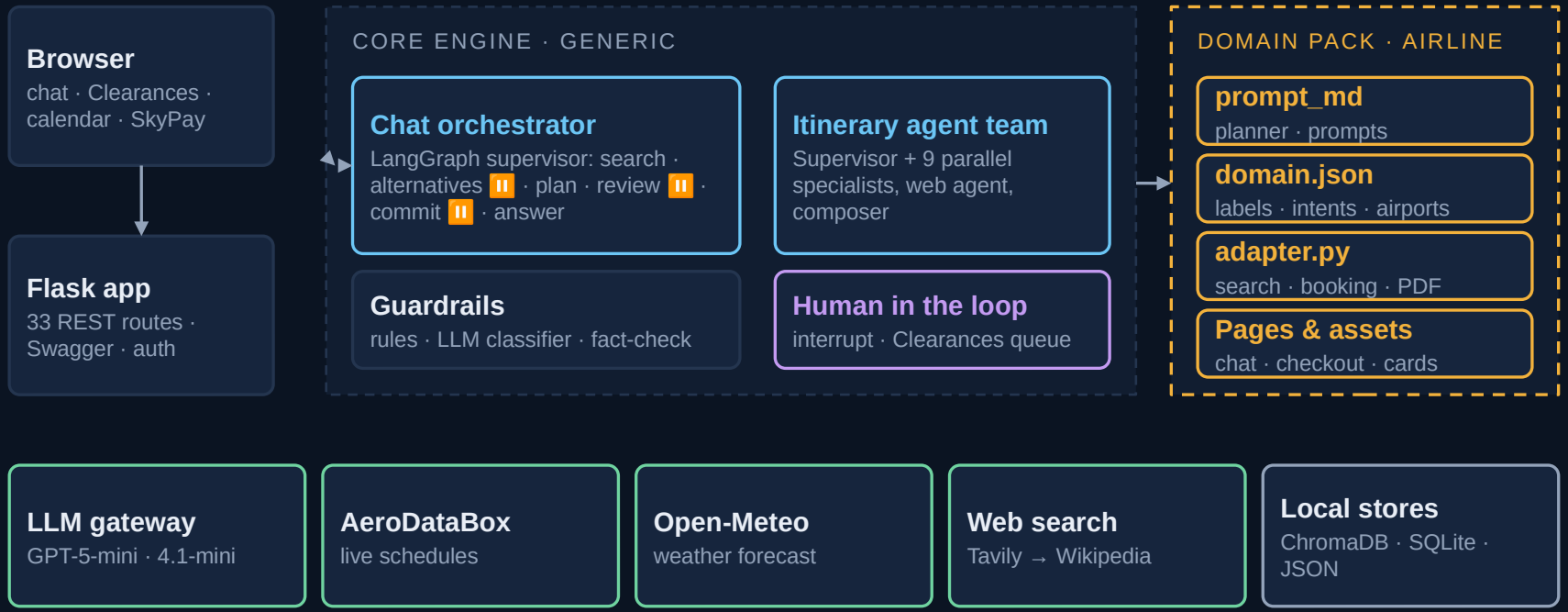
Frontend

server-rendered

Jinja templates · domain-aware
Vanilla JS · no framework
SkyPay checkout · demo payments

ARCHITECTURE

A generic engine with a plug-in domain pack



Below: external HTTPS APIs and local stores